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LAUDS FACTORIES

Open Call 2 | LAUDS Replication

Production Process Blueprint

DATA-RIDE

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Project coding LAUDSREP-01-19	*DATA-RIDE: Data and Advocacy for Transforming Active Riding & Infrastructure for a Democratic Environment*	
Deliverable title Production Process Blueprint: Data-Ride		**Due date** 01.10.2026
Hybrid Team entities **Lead partner:** re:edu GmbH & Co. KG **Partner 1: Laura Sobral** **Partner 2: Global Innovation Gathering (GIG)**		
Challenge Category Sustainable urban mobility	**Challenge Title** Citizen-led sensing and data-driven advocacy for safer cycling infrastructure	**Host LAUDS Factory** Université de Lorraine

1. Product and process Description

Please provide detailed information on the product or solution you have developed in your project for the LAUDS Open Call 2, including the elements detailed below.

Production Process

The technical development included adding a second Time-of-Flight (ToF) sensor to the senseBox setup for implementation during the Nancy campaign. Measuring lateral distances on both sides enabled exploration of the space available (corridors) around cyclists, complementing the analysis of potential overtaking events.

The sensing setup comprised a microcontroller (ESP32-S2), two ToF distance sensors, a particulate-matter sensor (SPS30), temperature and humidity sensors (HDC1080), an inertial measurement unit for motion and road-surface indicators (MPU6050), and a GPS data from the smartphone. Devices were mounted on participants' bicycles and connected to smartphones for recording and uploading data. Physical materials included sensor enclosures, mounting components, and a mounting prototype developed with local laboratory support. Online communication materials (Poster and Policy Recommendation) translated the findings into accessible visuals for public discussion.

The analysis used GPS position, speed, left- and right-side distances, particulate-matter measurements, and motion-based indicators of stopping and road-surface conditions. The processed dataset contained 193,328 observations across 357 trips, covering more than 1,000

kilometres and approximately 84 hours of valid recorded movement. Recordings were linked to OpenStreetMap street segments and combined with local datasets on cycling infrastructure, parking, and tree presence.

These inputs supported the production of a series of maps visualising the recorded cycling routes and different dimensions of the cycling experience. The maps included the spatial distribution of potential close-overtaking events, available lateral corridor widths during rides, road-surface roughness, and particulate-matter measurements. Additional spatial analyses combined the sensor recordings with cycling infrastructure and on-street parking data, making it possible to explore where constrained cycling space and potential close passes occurred in relation to different street and infrastructure conditions.

For the overtaking maps, potential events were identified by combining short-term patterns in left-side ToF distance measurements with the on-device machine-learning overtaking indicator. The analysis used two-second windows, filtered invalid distance measurements and stationary periods, removed repeated patterns potentially associated with parked cars, and merged nearby detections into individual events. These were mapped as potential overtaking events, rather than confirmed overtakes. Further local calibration and video-based validation would be needed to improve the reliability of the detection method

The local ecosystem involved the Métropole du Grand Nancy, the Lorraine Smart Cities Living Lab, EDEN, Atelier Dynamo,, Laboratoire Sauvage, volunteer cyclists, and workshop participants. Contributions ranged from hosting workshops and presentations to organising collections and supporting local outreach.

Resources included sensing devices, bicycles, smartphones, computing equipment, fabrication facilities, and digital services. Energy was required for charging, fabrication, data processing, and online services, although consumption was not quantified during the project.

Product Maturity

[Please compare the state of development and product maturity of the proposed solution at the start and end of your project (indicate TRL state) and detail the project's as well as the LAUDS Factories contribution to the development of the proposed solution (both from the host and the rest of the project, where relevant). Refer to modifications of the technical documentation.]

The project began with an existing senseBox:bike platform and a proposal to extend its functionality and apply it through a participatory campaign in Nancy. The support of LAUDS Factories was particularly important in enabling the team to move beyond technical development and test the system in a real-world setting with local cyclists.

By the end of the project, the system had been deployed with ten volunteers and supported a processed dataset, spatial analyses, public discussions, communication outputs, and policy

recommendations. The campaign also created opportunities to monitor participants' challenges in using the technology and to organise open sessions where sensor data could be compared with cyclists' lived experiences and perceptions.

This process helped reveal both the potential and the limitations of the approach. Maturity differs between components: field deployment was successfully demonstrated, while overtaking classification still requires further local calibration and validation. The participatory methodology was implemented and tested, showing the value of combining sensor measurements with local knowledge, while replication at a larger scale and policy uptake remain areas for further development.

Process and Blueprint

[Please provide and shortly discuss the final “blueprint” of the overall process (the document worked with SUPSI team; add it as an independent PDF file). Summarise how the project matured in additional aspects independent of the product itself: community building, new users, new business opportunities or models, relation to local authorities, etc. Refer to the open source business strategy if relevant.].

The final blueprint organises DATA-RIDE across four areas:

Blueprint area	Main contribution
Core Product	Citizen-generated cycling data, shared interpretation, and Policy Recommendation materials.
Implementation & Production	Prototype development, supported collection, and open documentation.
Ecosystem Partners	Actor mapping, partner-based interpretation, and policy recommendations.
Community & Customers	Accessible volunteer participation, public communication, and advocacy resources.

Beyond the product itself, the project matured through the creation of a local partner network, direct involvement of cyclists as participants and co-interpreters, and stronger connections with public authorities through the presentation and discussion of results. The Nancy campaign also helped clarify where technical support, calibration, documentation, and facilitation are needed for future deployments.

The sustainability approach therefore goes beyond maintaining the open-source hardware. It points toward a service model around an open-source core, including campaign setup, sensor adaptation and calibration, technical support, data processing and interpretation, participatory

workshops, and policy-oriented communication materials. This could support future collaboration with municipalities, research institutions, cycling organisations, or other local partners.

The blueprint also shows that replication requires more than copying the device. It requires a combination of open technology, technical support, local participation, contextual interpretation, partner coordination, and output such as the toolkit and policy recommendation materials. Community ownership and an active partner network are therefore as important to long-term sustainability as the technology itself.

Documentation

Add links to any online documentation (we recommend documenting the project's implementation using a Git Forge (GitHub, GitLab, Codeberg,...), following a template such as this one https://open-make.github.io/Hardware-template-guide/. Indicate here the licenses used.

Project documentation is being consolidated in the DATA-RIDE GitHub repository, including processing scripts, methodology, maps, toolkit and communication materials.

The code is licensed under the MIT License. The DATA-RIDE Blueprint is licensed under CC BY 4.0. Maps, toolkit and other project-authored communication materials are intended for release under CC BY-SA 4.0, with third-party content retaining its original licences and attribution requirements.

GitHub: https://github.com/reedu-reengineering-education/dataride-lauds

2. Impact

Please provide detailed information on your project's both already observed impacts, as well as further impacts that are expected beyond the project's conclusion, according to the elements detailed below.

Exploitation plan

[Please provide a plan or business model on how you intend to exploit the solution you have developed.]

DATA-RIDE's exploitation approach centres on the reuse of the tested technology and participatory methodology as an integrated service package. Potential applications include local cycling campaigns commissioned by municipalities or urban mobility departments, research collaborations, educational activities, and citizen-led initiatives supporting evidence-informed mobility discussions.

The service could combine:

- senseBox:bike devices and technical setup;
- local adaptation and an improved calibration and validation protocol;
- support for participatory data-collection campaigns;
- data processing, spatial analysis, maps and visualisations;
- local workshops to interpret the results together with cyclists and other stakeholders;
- communication and policy recommendation materials to support dialogue with local authorities.

Rather than offering the technology as a standalone product, the aim is therefore to provide a complete campaign and analysis service around an open-source technology.

The immediate priorities are to improve the calibration protocol, maintain accessible and reusable documentation, and consolidate the technical, analytical and participatory components into a clearly defined service package that can be replicated and adapted to different local contexts.

Sustainability and Community Involvement

[Please map the open, local, and ecological sustainability aspects of your project and developed solution, including how you have and will continue to integrate and involve the local ecosystem and communities.]

- Open sustainability: The project builds on open-source technology and aims to maintain openly accessible documentation, methodologies, data-processing workflows, maps. This allows the Nancy stakeholders and future communities and partners to adapt the approach without depending on a closed technological system.
- Local sustainability: The Nancy campaign involved local cyclists, cycling organisations, researchers, and local public actors throughout the process. Participants contributed not only by collecting data, but also by identifying relevant cycling issues and later interpreting the results based on their local knowledge and lived experience.
- Community involvement: Workshops created a feedback loop between participants and the project team. Feedback led to practical adaptations, including French-language materials and an alternative mounting solution for different bicycle configurations. The interpretation workshop also connected the collected evidence back to the issues raised by participants at the beginning of the campaign.
- Ecological sustainability: By supporting cycling advocacy and evidence-informed discussions about cycling infrastructure, DATA-RIDE contributes to broader efforts to strengthen active and low-carbon mobility.

Key exploitation

[Please indicate what were the most relevant activities you performed to generate lasting outcomes and why these are deemed particularly successful.]

The local discussions and the workshop, where sensor data were compared with cyclists' perceptions and experiences, generated not only maps but also four local directions for further exploration by the Nancy cycling community:

1. Promote safe overtaking and mutual respect.
2. Support new cyclists through peer learning.
3. Improve the quality and continuity of cycling infrastructure.
4. Continue developing citizen-generated mobility data.

Key Performance Indicators (KPIs)

[Please give metrics and explain how measurements were acquired for the KPIs of your project as defined in your proposal and during the project.]

KPI	Quantity	How it was measured
Participants engaged	10 volunteers	Registration / devices distributed during the campaign
Data-collection period	Feb-May	Sensor timestamps / campaign records
Recorded trips	357 trips	Trips identified in the processed sensor dataset
Distance recorded	1.234,74 km	GPS trajectories from the processed dataset
Cycling time recorded	84 h	Sensor timestamps / trip duration
Sensor data points	193.328	Processed sensor dataset
Participatory workshops	3	(February) launch workshop, Exploratory workshop + (June) data Interpretation workshop
Public dissemination events	2	Fête du Vélo public presentation and Online presentation at Lorraine University
Types of spatial analysis produced	9	Standing Points, Routes,Bike path conditions, Speed, Overtaking events and distance, corridor/parking lot and

		corridors, particulate matter/trees, pollution
Local adaptations implemented	2	French translation + Prototype of an alternative bicycle mounting system
Open outputs produced	Dataset, methodology, maps, policy recommendations, blueprint, communication materials	Final project documentation / repositories

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